

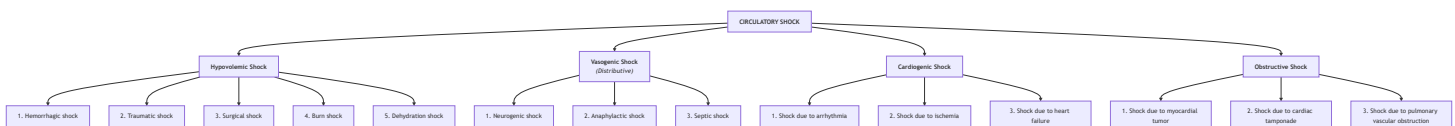
TYPES AND CAUSES OF CIRCULATORY SHOCK

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- Circulatory shock is classified into **4 major types**:
 - a. **Hypovolemic Shock** (Due to ↓ decreased blood volume)
 - b. **Vasogenic / Distributive Shock** (Due to ↑ increased vascular capacity)
 - c. **Cardiogenic Shock** (Due to cardiac disease)
 - d. **Obstructive Shock** (Due to obstruction of blood flow)

CLASSIFICATION OVERVIEW



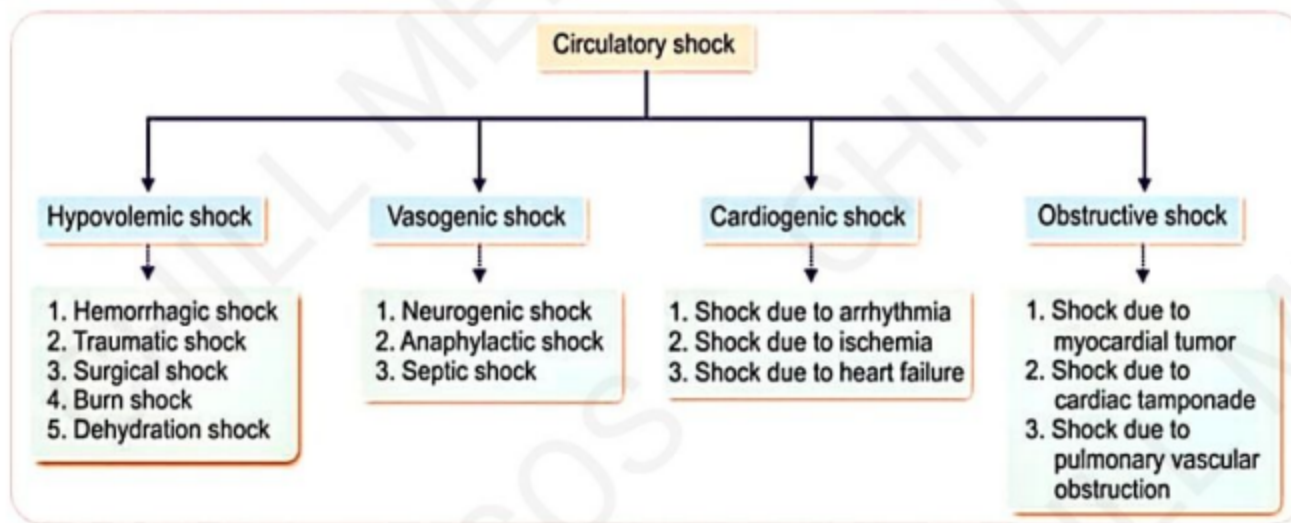


FIGURE 116.4: Different types of circulatory shock

1. HYPOVOLEMIC SHOCK (COLD SHOCK)

- **Definition** : Shock caused by a critical reduction in blood volume.
- **Threshold** : Occurs with an acute loss of at least 10% – 15% of total blood volume.
(Loss < 10% does not cause shock due to immediate compensatory mechanisms).
- **Key Manifestations** :—
 - ↓ Cardiac output
 - Low blood pressure (Hypotension)
 - Thin, thready, rapid pulse
 - Pale, cold, clammy skin
 - ↑ Increase in respiratory rate (Tachypnea)

Pathological Subtypes of Hypovolemic Shock:

- **a. Hemorrhagic Shock** :—
 - **Acute Hemorrhage** : Massive blood loss in accidents/trauma.
 - **Chronic Hemorrhage** : Slow, persistent blood loss (e.g., bleeding peptic ulcers).
- **b. Traumatic Shock** :—
 - Caused by severe injury or wounds from external force (battlefields, major road accidents).
 - **Crush Syndrome** : Acute renal failure developing when a limb is crushed/compressed; myoglobin and toxic metabolites released from injured muscle damage renal tubular cells, leading to tubular degeneration.

- **Reperfusion Injury** : Tissue dysfunction and cellular injury caused when blood supply returns to tissue after a period of ischemia.
- **c. Surgical Shock** :—
- Due to internal/external blood loss and fluid loss/dehydration during or after major operative procedures.
- **d. Burn Shock** :—

Extensive burn injury → Massive loss of plasma through burnt surface



↓ ECF volume & plasma volume → **Hemoconcentration** → ↓ Cerebral blood flow

- **e. Dehydration Shock** :—
- Caused by severe decrease in total body water content (*e.g., cholera, severe vomiting, prolonged diarrhea*).

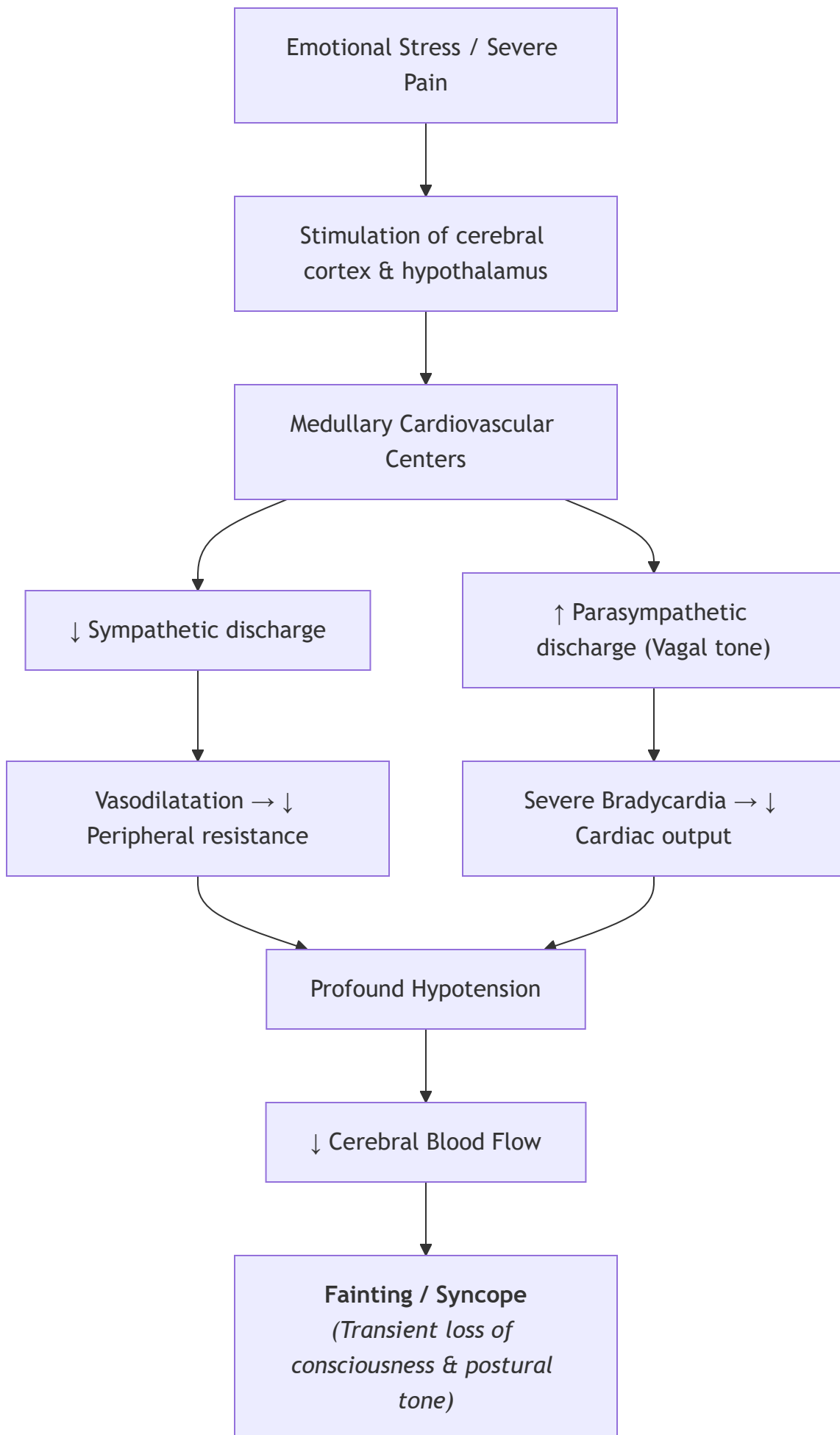
2. VASOGENIC SHOCK (DISTRIBUTIVE / LOW RESISTANCE SHOCK)

- **Definition** : Inadequate tissue perfusion caused by widespread, abnormal **vasodilation** that creates an enormous increase in the vascular bed capacity relative to normal blood volume.

A. Neurogenic Shock

- Extensive vasodilation caused by the **sudden loss of vasomotor tone**.
- **Etiology** : Brain ischemia, general anesthesia, spinal anesthesia, extreme emotional trauma.

Mechanism of Vasovagal Syncope (Emotional Fainting):



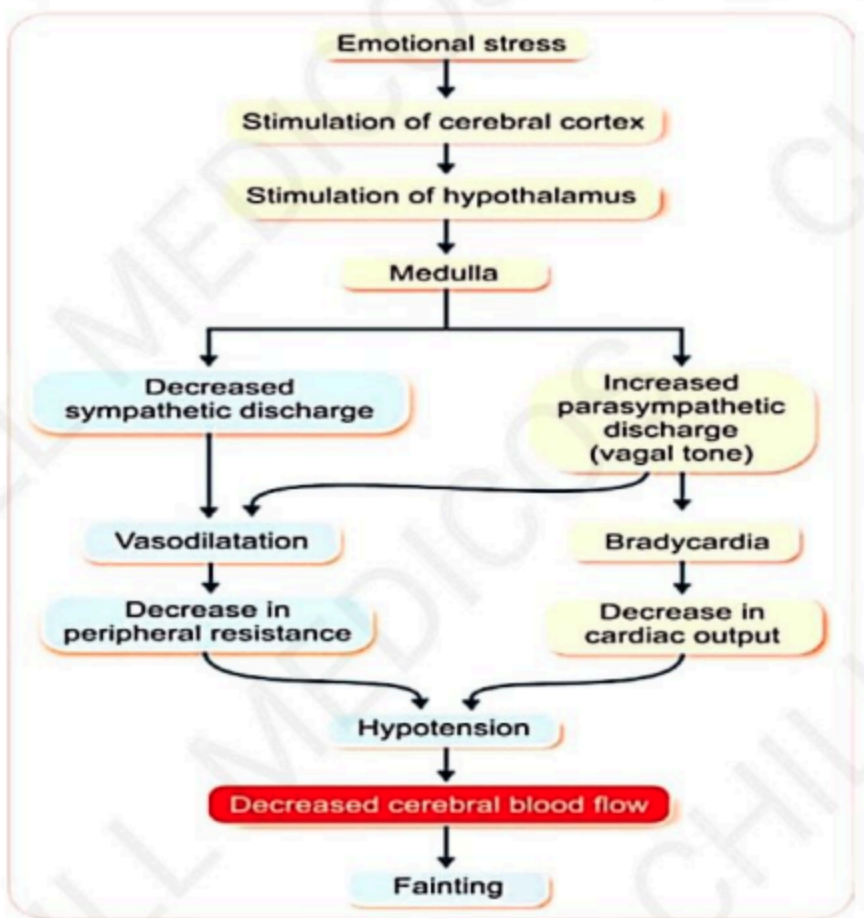


FIGURE 116.5: Schematic representation of vasovagal syncope

B. Anaphylactic Shock

- Severe, life-threatening **systemic allergic reaction** triggered by exposure to an antigen/foreign protein to which the individual was previously sensitized (*massive release of histamine and leukotrienes*).

C. Septic Shock (Endotoxic Shock)

- Severe pathological state characterized by the presence of pathogenic microorganisms or their toxins (**endotoxins**) in blood or tissues (*blood poisoning*).
- **Common Causes** : Peritoneal infections, severe bacterial skin infections (*Staphylococcus / Streptococcus*).
- *Has features overlapping vasogenic, cardiogenic, and hypovolemic shock.*

3. CARDIOGENIC SHOCK

- Shock originating from **primary intrinsic pump failure** of the heart.
- **Etiology :—**
 - a. Congestive heart failure / advanced cardiac disease
 - b. Severe arrhythmias (leading to ↓ cardiac output)
 - c. Depressed myocardial contractility due to acute ischemia / infarction

4. OBSTRUCTIVE SHOCK

- Shock caused by physical or mechanical **obstruction to blood flow** through the central cardiovascular circuit.
- **Etiology :—**
 - a. Intracardiac tumors (*e.g., atrial myxoma*)
 - b. **Cardiac Tamponade** (*compression of heart by fluid in pericardial sac*)
 - c. **Pulmonary Embolism** (*massive obstruction of pulmonary vessels*)

GENERAL TREATMENT MODALITIES FOR SHOCK

- a. **Blood Transfusion** : Essential in acute hemorrhagic shock.
- b. **Plasma Transfusion** : Indicated in burn shock and severe hypoproteinemia.
- c. **Administration of Plasma Substitutes** : Dextran, gelatin solutions, or balanced crystalloids/colloids.
- d. **Sympathomimetic Drugs** : Inotropic agents and vasopressors (*e.g., Norepinephrine, Dopamine, Epinephrine*) to restore vascular tone and cardiac contractility.
- e. **Administration of Glucocorticoids** : High-dose steroids to stabilize lysosomal membranes, combat endotoxins, and reduce capillary permeability.
- f. **Oxygen Therapy** : Corrects systemic and tissue hypoxia.
- g. **Postural Adjustment** : Placing the patient in Trendelenburg position (head low, legs elevated) to improve venous return to the brain and heart.